

EN.553.740 Machine Learning 1

Jonathan Ma

September 26, 2026

Contents

0.1	Introduction	1
0.1.1	Learning Problems	2
0.2	Mathematical Review	2
0.2.1	Linear Algebra	2
0.2.2	Calculus	9
0.2.3	Probability	15

0.1 Introduction

A good way to think of machine learning is instead of a computer taking data and a program to produce output, it takes data and output and attempts to produce a program, ie "learn" a function.

We use it for tasks that are too complex to program, like driving or large scale data analysis. A bit of history of "learning":

- 1959: Arthur Samuel (IBM) created a program to play checkers, introducing rule-based learning.
- 1960: Frank Rosenblatt introduced a linear classifier based on brain neurons, called a Perceptron.
- 1969: Marvin Minsky showed limitations of the perceptron, halting development for 20 years
- 1980: First workshop in machine learning
- 1986: Multilayer networks and backpropagation introduced
- 1990: Vapnik and Chervonenkis introduce kernels and SVM, and termed "Statistical Learning"
- 2002: Rise of graphical models
- 2010: Deep learning takes over
- 2016: ML hits record size with NIPS and ICML doubling in size

Machine learning has been used for a variety of tasks, like stock market prediction, decoding thoughts from brain scans, and self-driving cars.

0.1.1 Learning Problems

Statistical Learning falls under five rough fields:

- Classification (Assign a category)
- Regression (predict a real value)
- Ranking (order items)
- Clustering (item partitioning, like image segmentation)
- Dimensionality Reduction/Manifold Learning (Computer vision)

In a broader sense, there is supervised learning (Classification/regression) and unsupervised learning (Clustering, dimensionality reduction, density estimation) and things like semi-supervised learning, active learning, reinforcement learning, etc.

Supervised Learning For a given problem, we learn a function $f : X \rightarrow Y$ such that $x \in \mathcal{X}$, $y \in \mathcal{Y}$. Classification deals with discrete labels while regression deals with continuous functions.

Unsupervised Learning For a given problem, we learn a function $f : X \rightarrow Y$ such that $x \in \mathcal{X}$, $f(x) \in \mathbb{R}$. Usually embedding (for text), clustering, or density estimation.

0.2 Mathematical Review

Algorithms are built from sets, functions, vectors, matrices, norms, and derivatives, so we introduce common notation here. Consider a student success dataset with the target being quiz score, and features as study time and prior score.

0.2.1 Linear Algebra

0.2.1.1 Sets and Power Sets

Let $A = \{1, 2, 3, 4, 5\}$ be the set of students in the toy dataset. The *power set* is $\mathcal{P}(A) = \{B : B \subseteq A\}$. If $|A| < \infty$, then $|\mathcal{P}(A)| = 2^{|A|}$. A subset $B \subseteq A$ may represent a mini-batch, validation set, or a selected group of samples.

0.2.1.2 Functions

For sets A, B , write $B^A = \{f : A \rightarrow B\}$. if $B = \mathbb{R}$, then $\mathbb{R}^A$ is the set of real-valued functions on A . If $|A| < \infty$, then $\mathbb{R}^A \simeq \mathbb{R}^{|A|}$ as vector spaces. For our toy example:

$$x_1 : A \rightarrow \mathbb{R} \quad x_2 : A \rightarrow \mathbb{R} \quad y : A \rightarrow \mathbb{R}$$

For a subset $C \subseteq A$, define an *indicator function*:

$$\mathbb{1}_C : A \rightarrow \{0, 1\}, \quad \mathbb{1}_C(i) = \begin{cases} 1, & i \in C, \\ 0, & i \notin C \end{cases}$$

Indicators are used to encode events, groups, labels, masks, etc.

0.2.1.3 Vectors

We use a column convention for vectors. For each student i , write the feature vector as:

$$x_i = \begin{bmatrix} x_i^{(1)} \\ x_i^{(2)} \end{bmatrix} \in \mathbb{R}^2$$

Standard vectors that are useful include the zero vector (0_d), unit vector ($\mathbb{1}_d$), and identity matrix ($I_{\mathbb{R}^d}$):

$$0_d = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} \quad \mathbb{1}_d = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \quad I_{\mathbb{R}^d} = \begin{bmatrix} 1 & \cdots & 0 \\ & \ddots & \\ 0 & \cdots & 1 \end{bmatrix}$$

0.2.1.4 ℓ^p norms

For $p \geq 1$ and $x \in \mathbb{R}^d$, a p -norm is defined as

$$\|x\|_p = (|x^{(1)}|^p + \cdots + |x^{(d)}|^p)^{\frac{1}{p}}$$

. Special cases include:

$$\ell_1 = \sum_j |x^{(j)}| \quad \ell_2 = \sqrt{\sum_j (x^{(j)})^2} \quad \ell_\infty = \max_j |x^{(j)}|$$

ℓ^2 is called the euclidean norm. We require $p \geq 1$ because the triangle inequality holding is essential for a norm, and only holds for $p \geq 1$:

$$\|x + y\|_p \leq \|x\|_p + \|y\|_p$$

$$\text{when } p = 1; |x| + |y| \leq 1$$

$$\text{when } p = \infty; \max\{|x|, |y|\} \leq 1 \iff |x| \leq 1,$$

A good visualization of norms is shown below:

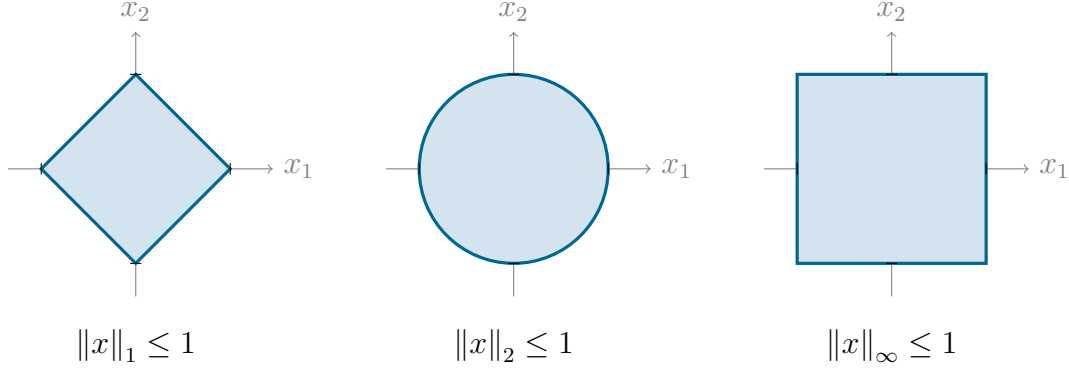


Figure 1: Unit balls in $\mathbb{R}^2$. Their boundaries consist of vectors with norm equal to one.

Operator Norms For $A \in \mathcal{M}_{m,d}$, $\|A\|_{op} := \max\{\|Ax\|_2 : x \in \mathbb{R}^d, \|x\|_2 = 1\}$. $\|A\|_{op}$ is the largest stretching factor of the linear map $x \mapsto Ax$. Alternatively, we can derive a similar definition:

$$\begin{aligned}
 \|Ax\|_2^2 &= \langle Ax, Ax \rangle \\
 \langle Ax, Ax \rangle &= (Ax)^T Ax = x^T A^T Ax \\
 \|A\|_{op}^2 &= \max_{\|x\|_2=1} x^T A^T Ax
 \end{aligned}$$

Operator norms quantify how much perturbations can be amplified.

0.2.1.5 Cauchy-Schwarz Inequality

For $x, y \in \mathbb{R}^d$, $|\langle x, y \rangle| \leq \|x\|_p \|y\|_q$ with equality iff $x = \alpha y$ for some scalar α and $\frac{1}{p} + \frac{1}{q} = 1$.

Proof sketch: For any $\alpha \in \mathbb{R}$,

$$0 \leq \|x - \alpha y\|_2^2 = \|x\|_2^2 - 2\alpha \langle x, y \rangle + \alpha^2 \|y\|_2^2$$

Choosing the best α gives the inequality:

$$\begin{aligned}
 &\forall \alpha \in \mathbb{R}, \quad \|x - \alpha y\|_2^2 \geq 0 \\
 &\iff \langle x - \alpha y, x - \alpha y \rangle \geq 0 \\
 &\iff \langle x, x \rangle - 2\alpha \langle x, y \rangle + \alpha^2 \langle y, y \rangle \geq 0 \\
 &\iff \|x\|_2^2 - 2\alpha \langle x, y \rangle + \alpha^2 \|y\|_2^2 \geq 0
 \end{aligned}$$

Without loss of generality, assume $\|y\|_2 \geq 0$ and take $\alpha = \frac{\langle x, y \rangle}{\|y\|_2^2}$:

$$\begin{aligned}
 \implies \|x\|_2^2 - \frac{2 \langle x, y \rangle \|x\|_2}{\|y\|_2} + \|x\|_2 &\geq 0 \\
 \|x\|_2 \|y\|_2 &\geq \langle x, y \rangle
 \end{aligned}$$

Equality means $x - \alpha^* y = 0$ so $x = \alpha^* y$.

Cauchy-Schwarz is used in ML to control linear scores, since most models are about estimating weights. A linear model estimates $f_w(x) = \langle w, x \rangle$. Cauchy-Schwarz gives

$$|f_w(x)| = |\langle w, x \rangle| \leq \|w\|_2 \|x\|_2$$

This is saying if features have a bounded norm, and weights have a bounded norm, scores cannot explode. This is why norm constraints and regularization appear throughout learning problems.

The inner product cannot be larger than length times length

We now look at a particle sensor example. Suppose independent sensors $X_1, X_2, \dots, X_n$ estimate the same mean μ :

$$\mathbb{E}(X_i) = \mu \quad \text{Var}(X_i) = \sigma_i^2$$

Among weighted averages

$$\hat{\mu} = \sum_{i=1}^n w_i X_i$$

which weights should we use, keeping $\hat{\mu}$ unbiased? Compute:

$$\mathbb{E}(\hat{\mu}) = \mathbb{E}\left[\sum_i w_i X_i\right] = \sum_i w_i \mathbb{E}X_i = \sum_i w_i \mu = \mu$$

Thus

$$\mathbb{E}(\hat{\mu}) = \mu \iff \sum_i w_i = 1$$

Because X_i are independent,

$$\text{Var}(\hat{\mu}) = \text{Var}\left(\sum_i w_i X_i\right) = \sum_i w_i^2 \text{Var}(X_i) = \sum_i w_i^2 \sigma_i^2$$

The optimization problem we obtain is therefor minimize $\text{Var}(\hat{\mu})$ s.t. $\sum_i w_i = 1$ since the sum of weights is one,

$$1 = \left(\sum_i w_i\right)^2 = \left(\sum_i (w_i \sigma_i) \frac{1}{\sigma_i}\right)^2$$

Now by Cauchy-Schwarz,

$$1 = \left(\sum_i w_i\right)^2 \leq \left(\sum_i w_i^2 \sigma_i^2\right) \left(\sum_i \frac{1}{\sigma_i^2}\right)$$

Recall $|\langle x, y \rangle| \leq \|x\|_2 \|y\|_2$. Therefore

$$\text{Var}(\hat{\mu}) = \sum_i w_i^2 \sigma_i^2 \geq \left(\sum_i \sigma_i^{-2}\right)^{-1}$$

This makes it so the elements of X and y are $w_i \sigma_i$ and $1/\sigma_i$ respectively.

Equality in Cauchy-Schwarz holds when

$$w_i \sigma_i \propto \sigma_i^{-1} \implies w_i \propto \frac{1}{\sigma_i^2}$$

so we should normalize weights so they sum to one:

$$\frac{\frac{1}{\sigma_i^2}}{\sum_{j=1}^n \frac{1}{\sigma_j^2}}$$

meaning lower variance sensor implies larger weight.

0.2.1.6 Sparsity and Lasso

Sparsity is roughly defined as what happens when $p \rightarrow 0$. For a nonzero scalar a :

$$\lim_{p \rightarrow 0} \sum_j |x^{(j)}|^p = \sum_j \lim_{p \rightarrow 0} |x^{(j)}|^p \because \lim_{p \rightarrow 0} |a|^p = 1 \text{ (imagine } \begin{cases} 0, x^{(j)} = 0 \\ 1, x^{(j)} \neq 0 \end{cases})}$$

For $a = 0$, $|0|^p = 0$. Therefore informally,

$$\lim_{p \rightarrow 0} \sum_j |x^{(j)}|^p = \{j : x^{(j)} \neq 0\} = \|x\|_0$$

The quantity $\|x\|_0$ counts the number of nonzero entries, but it is not a norm and optimization with it is often difficult. Common times in ML, we replace $\|x\|_0$ with $\|x\|_1$ to encourage sparsity while keeping convexity

0.2.1.7 Matrices

In ML, rows often represent samples and columns represent features. we say $\mathcal{M}_{m,d} = \mathcal{M}_{m,d}(\mathbb{R}) = \{m \times d \text{ real matrices}\}$. Additionally $\mathcal{M}_d = \mathcal{M}_{d,d}$. Notation may differ, so remember its always (across, down) ie (i,j)/(r,c)/(m,d)

By this definition, a linear prediction vector is $X_w = [\langle x_1, w \rangle, \langle x_2, w \rangle, \dots, \langle x_i, w \rangle]^T$.

Symmetric and Positive Semidefinite Matrices A symmetric matrix is defined as

$$S_d = \{A \in \mathcal{M}_d : A = A^T\}$$

A positive semidefinite matrix (PSD) is defined as

$$S_d^+ = \{A \in S_d : x^T A x \geq 0 \quad \forall \quad x \in \mathbb{R}^d\}$$

These usually appear as covariance and correlation matrices, and Hessians of convex quadratic objective functions. A positive definite matrix is defined as

$$S_d^{++} = \{A \in S_d : x^T A x > 0 \quad \forall \quad x \neq 0\}$$

This usually comes up when doing ridge regressions and finding hessians with a parameter greater than 0.

Bilinear Forms A bilinear form is a map $a : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ that is linear in each variable. If a is symmetric (ie $a(x, y) = a(y, x)$) then $a(x, y) = x^T A y$ for some symmetric matrix $A \in S_d$.

Inner products are PSD:

$$\begin{aligned} a(x, y) &= \langle x, y \rangle = x^T y \\ a(x, y) &= \langle x, y \rangle = x^T I_d y \\ a(x, x) &= x^T x = \|x\|_2^2 \geq 0 \end{aligned}$$

Orthogonal Matrices An Orthogonal transformation matrix O is defined as

$$O_{m,d} = \{A \in \mathcal{M}_{m,d} : AA^T = I_m\}$$

and for $Q \in O_d$, $Q^T Q = Q Q^T = I_d$. The key idea is orthogonal transformations preserve euclidean geometry:

$$\|Qx\|_2^2 = (Qx)^T (Qx) = x^T Q^T Q x = x^T x = \|x\|_2^2$$

Similarly

$$\langle Qx, Qy \rangle = x^T Q^T Q y = \text{inner } xy$$

Rotations preserve distances and dot products, which matter for PCA, kernels, and preprocessing

0.2.1.8 Open and Closed Balls

We introduce some basic topology. For $x \in \mathbb{R}^d, r > 0$: A ball is open if

$$B(x, r) = \{y \in \mathbb{R}^d : \|y - x\|_2 < r\}$$

A ball includes the boundary if

$$\bar{B}(x, r) = \{y \in \mathbb{R}^d : \|y - x\|_2 \leq r\}$$

A set $U \subset \mathbb{R}^d$ is open if

$$\forall x \in U, \quad \exists r > 0, \quad B(x, r) \subset U$$

A set is closed if its complement is open. The intuition is that inside an open set, every point has a small safety ball around it.

For $A \subset \mathbb{R}^d$, the interior ($\text{int}(A)$) is the largest open set containing A , and the closure $\text{cl}(A)$ is the smallest closed set containing A . We can also think of this as the closure including the boundary whereas the interior doesn't.

Compactness : A set $K \subset \mathbb{R}^d$ is compact if every sequence of points in K contains a subsequence that converges to a point in K . Take for instance $(0,1)$ vs $[0,1]$: The sequence $x_n = n^{-1}$ lies in $(0,1)$ and converges to 0, but $0 \notin (0,1)$, so $(0,1)$ is not compact. For $[0,1]$, limits of convergent subsequences stay inside the set.

Heine-Borel Theorem A finite-dimensional Euclidean space:

$$K \subset \mathbb{R}^d \text{ compact} \iff K \text{ is closed and bounded.}$$

Bounded means $\exists R > 0$ s.t. $K \subset B(0, R)$. This is special to finite dimensional euclidean spaces, not other spaces.

Weierstrass Extreme Value Theorem If $K \subset \mathbb{R}^d$ is compact and $f : K \rightarrow \mathbb{R}$ is continuous, then f attains its minimum and maximum on K . Put otherways,

$$\exists x_{min}, x_{max} \in K \text{ s.t. } f(x_{min}) = \min_{x \in K} f(x), \quad f(x_{max}) = \max_{x \in K} f(x)$$

In optimization, we often write $\inf_{w \in K} \ell(w)$. Weierstrass tells us when this minimum is actually achieved. Basically saying that on compact parameter sets, continuous objectives have optimizers. We look at an example:

Look at constrained regression. Consider minimizing least squares over a compact ball

$$K_R = \{w \in \mathbb{R}^d : \|w\|_2 \leq R\}$$

The objective is

$$\ell(w) = \frac{1}{2n} \|Xw - y\|_2^2$$

We can prove this by showing $\ell(w)$ is continuous in w . Then show that K_R is closed and bounded in $\mathbb{R}^d$. By Heine-Borel, K_R is compact, so by Weierstrass, $\exists \hat{w} \in K_R$ s.t. $\ell(\hat{w}) = \min_{w \in K_R} \ell(w)$.

0.2.1.9 Metric Spaces

A metric space is a set $\mathcal{B}$ equipped with a distance $\rho : \mathcal{B} \times \mathcal{B} \rightarrow [0, \infty)$ satisfying $\forall x, y, z \in \mathcal{B}$:

$$\rho(x, y) = 0 \iff x = y$$

$$\rho(x, y) = \rho(y, x)$$

$$\rho(x, z) \leq \rho(x, y) + \rho(y, z)$$

For $\mathcal{B} \in \mathbb{R}$ and $\rho : \mathbb{R} \times \mathbb{R} : \rho(x, y) = |x - y|$.

For $\mathcal{B} \in \mathbb{R}^d$ and $\rho : \mathbb{R}^d \times \mathbb{R}^d : \rho(x, y) = \|x - y\|_2$.

Metrics formalize dissimilarity between data points. This could be distances between two feature vectors, probability distributions, or functions/signals. Open, Closed, and compactness still make sense in metric spaces, but Heine-Borel may fail. In $\mathbb{R}$, closed and bounded iff compact, but in a general metric space, it does not imply compactness

0.2.2 Calculus

Calculus Notation Map

Notation	Meaning in these notes
$x, y \in \mathbb{R}$	one-dimensional points, often with $x < y$
$x \in \mathbb{R}^d$	deterministic vector or input point
$e_1, \dots, e_d$	canonical basis of $\mathbb{R}^d$: $e_1 = (1, 0, \dots)$, $e_2 = (0, 1, \dots)$
∂f	derivative in one dimension
$\partial_j f$	partial derivative along e_j
$df(x)$	differential or Jacobian matrix at x
$\nabla \cdot f$	divergence, when $f: \mathbb{R}^d \rightarrow \mathbb{R}^d$
$\nabla^2 f$	Hessian, for scalar-valued f
Δf	Laplacian, $\text{tr}(\nabla^2 f)$

Probability Notation Map

Notation	Meaning in these notes
Ω	sample space
$A \subseteq \Omega$	event
$\mathbb{P}$	probability function or probability measure
$(\Omega, \mathbb{P})$	probability space
X, Y	random variables (capital letters)
x, y	realizations or values of random variables
R_X, R_Y	possible values of X and Y
$[X = x]$	event $\{\omega \in \Omega : X(\omega) = x\}$
$\mathbb{E}[X]$	expectation of X
$\mathbb{E}[Y \mid X = x]$	conditional expectation of Y given $X = x$

A model is often a function $f(x)$ and a loss is often a function $\ell(w)$ of parameters w . Optimization therefore asks how does $\ell(w)$ change after w is perturbed, or, if we move from w to $w + h$, can we predict whether loss goes down?

Intervals Assume $x, y \in \mathbb{R}; x > y$. The line segment from x to y is written by $(1 - t)x + ty$. Define the intervals as follows:

$$[x, y] := \{(1 - t)x + ty : 0 \leq t \leq 1\}$$

$$[x, y) := \{(1 - t)x + ty : 0 \leq t < 1\}$$

$$(x, y] := \{(1 - t)x + ty : 0 < t \leq 1\}$$

$$(x, y) := \{(1 - t)x + ty : 0 < t < 1\}$$

The same line segment formula will appear in convexity, Jensen's Inequality, and Taylor expansion along a direction.

0.2.2.1 1D Derivatives

Let $f : I \subset \mathbb{R} \rightarrow \mathbb{R}$, $t \mapsto f(t) \in \mathbb{R}$. The derivative of f at $t_0 \in I$ is

$$\partial f(t_0) = \partial f|_{t=t_0} := \lim_{h \rightarrow 0} \frac{f(t_0 + h) - f(t_0)}{h}$$

when this limit exists. For higher-order derivatives:

$$\partial^k f := \partial(\cdots \partial(\partial f)), \quad k \geq 0, \quad k \in \mathbb{Z}^+$$

with the convention $\partial^0 f := f$. The derivative is the local slope of the function at t_0 .

0.2.2.2 Higher-Dimensional Derivatives

Now move to $d > 1$, and let $f : U \subset \mathbb{R}^d \rightarrow \mathbb{R}^m$ and U is *open* (meaning we can move a little around x without immediately leaving the domain).

0.2.2.3 Partial Derivates

Let $e_1, \dots, e_d$ be the canonical basis of $\mathbb{R}^d$. For $1 \leq j \leq d$:

$$\partial_j f(x) := \partial_t(f(x + te_j))|_{t=0} = \lim_{t \rightarrow 0} \frac{f(x + te_j) - f(x)}{t}$$

This means that e_j changes only coordinate j and $x + te_j$ means move from x only along coordinate j . The partial derivative measures local change in that coordinate direction.

For example, let $f(\theta) = f(x, y) = x^2 + 3xy$. Then $\partial_x f(\theta) = 2x + 3y$ and $\partial_y f(x) = 3y$, and at $\theta = (1, 2)^T$, $\partial_x f(\theta) = 8$ and $\partial_y f(x) = 3$

0.2.2.4 Directional Derivatives

Partial derivatives measure change along the coordinate directions. More generally, for a direction $h \in \mathbb{R}^d$, the *directional derivative* of f at x in the direction h is

$$\partial_t f(x + th)|_{t=0}.$$

A partial derivative is the special case $h = e_j$, whereas a directional derivative allows any $h \in \mathbb{R}^d$.

In machine learning, if x is an input image or feature vector and h is a small perturbation, the directional derivative describes how the model output changes under that perturbation.

0.2.2.5 The Differential

Let $f: U \subset \mathbb{R}^d \rightarrow \mathbb{R}^m$, with component functions $f^{(1)}, \dots, f^{(m)}$. The *differential* of f at x is

$$df(x) := \left(\partial_j f^{(i)}(x) \right)_{1 \leq i \leq m, 1 \leq j \leq d}.$$

Equivalently,

$$df(x) = \begin{pmatrix} \partial_1 f^{(1)}(x) & \cdots & \partial_d f^{(1)}(x) \\ \vdots & \ddots & \vdots \\ \partial_1 f^{(m)}(x) & \cdots & \partial_d f^{(m)}(x) \end{pmatrix} \in \mathbb{R}^{m \times d}.$$

Thus, row i contains the partial derivatives of the output component $f^{(i)}$, while column j contains the partial derivatives with respect to the input coordinate x_j .

Example 0.2.1 (Computing $df(x)h$). Consider the scalar-valued function

$$f(x^{(1)}, x^{(2)}) = (x^{(1)})^2 + 3x^{(1)}x^{(2)}.$$

Its differential is the row vector

$$df(x) = \begin{pmatrix} \partial_1 f(x) & \partial_2 f(x) \end{pmatrix} = \begin{pmatrix} 2x^{(1)} + 3x^{(2)} & 3x^{(1)} \end{pmatrix}.$$

At $x = (1, 2)^T$,

$$df(x) = \begin{pmatrix} 8 & 3 \end{pmatrix}.$$

Therefore, if $h = (1, -1)^T$, then

$$df(x)h = \begin{pmatrix} 8 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 5.$$

0.2.2.6 Product Rule

Let $f, g: U \rightarrow \mathbb{R}$. Then

$$d(fg)(x) = f(x) dg(x) + g(x) df(x).$$

The product rule says that the first-order change of a product has two contributions: one from changing f and one from changing g .

0.2.2.7 Chain Rule

Let

$$f: U \subset \mathbb{R}^d \rightarrow \mathbb{R}^m, \quad g: V \subset \mathbb{R}^b \rightarrow U.$$

The condition $g(V) \subseteq U$ ensures that $f(g(x))$ is defined for every $x \in V$. The composition satisfies $f \circ g: V \rightarrow \mathbb{R}^m$, and the chain rule is

$$d(f \circ g)(x) = df(g(x)) dg(x).$$

The dimensions are compatible because

$$\underbrace{df(g(x))}_{m \times d} \underbrace{dg(x)}_{d \times b} = \underbrace{d(f \circ g)(x)}_{m \times b}.$$

Modern machine-learning models are compositions of functions:

$$x \mapsto h_1(x) \mapsto h_2(h_1(x)) \mapsto \cdots \mapsto f(x).$$

The chain rule explains how derivatives propagate through such a composition. In particular, backpropagation in a neural network is repeated application of the chain rule.

0.2.2.8 Divergence

When $d = m$, a function $f: U \subset \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a vector field. Its *divergence* is

$$\nabla \cdot f(x) := \text{tr}(df(x)).$$

Writing

$$f(x) = \begin{pmatrix} f^{(1)}(x) \\ \vdots \\ f^{(d)}(x) \end{pmatrix},$$

the divergence is the sum of the diagonal entries of the $d \times d$ differential:

$$\nabla \cdot f(x) = \partial_1 f^{(1)}(x) + \cdots + \partial_d f^{(d)}(x).$$

Example 0.2.2 (Computing divergence). Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be

$$f(x) = \begin{pmatrix} (x^{(1)})^2 \\ x^{(1)}x^{(2)} \end{pmatrix}.$$

Then

$$df(x) = \begin{pmatrix} 2x^{(1)} & 0 \\ x^{(2)} & x^{(1)} \end{pmatrix}.$$

Therefore,

$$\nabla \cdot f(x) = \text{tr}(df(x)) = 2x^{(1)} + x^{(1)} = 3x^{(1)}.$$

Divergence appears in continuous normalizing flows, probability flows, and differential equations for densities.

0.2.2.9 Determinant and Its Derivatives

Let

$$A = \begin{bmatrix} a_1 & \cdots & a_d \end{bmatrix} \in \mathcal{M}_d,$$

where a_i is the i th column of A . The determinant

$$\det: \mathcal{M}_d \rightarrow \mathbb{R}$$

can be viewed as a d -linear form in the columns:

$$\delta(a_1, \dots, a_d) = \det(A).$$

It is characterized by the following properties:

- If $a_i = a_j$ for $i \neq j$, then $\delta(a_1, \dots, a_d) = 0$.
- Swapping two columns multiplies δ by -1 .
- $\delta(e_1, \dots, e_d) = 1$.

Let a_{ij} denote the entry in row i and column j of A , and let $A^{(i,j)}$ be the matrix obtained by deleting row i and column j . The entrywise derivative of the determinant is

$$\partial_{a_{ij}} \det(A) = (-1)^{i+j} \det(A^{(i,j)}).$$

The quantity $(-1)^{i+j} \det(A^{(i,j)})$ is the *cofactor* corresponding to the entry a_{ij} .

Proof. Expand the determinant along column j :

$$\det(A) = \sum_{r=1}^d (-1)^{r+j} a_{rj} \det(A^{(r,j)}).$$

Differentiating with respect to a_{ij} gives

$$\begin{aligned} \partial_{a_{ij}} \det(A) &= \partial_{a_{ij}} \left[\sum_{r=1}^d (-1)^{r+j} a_{rj} \det(A^{(r,j)}) \right] \\ &= (-1)^{i+j} \det(A^{(i,j)}). \end{aligned}$$

Only the term with $r = i$ contains a_{ij} directly. □

0.2.2.10 Higher-Order Derivatives

Higher-order derivatives are defined iteratively from first-order derivatives. For $f: U \rightarrow \mathbb{R}^m$,

$$\partial_{i_k} \cdots \partial_{i_1} f: U \rightarrow \mathbb{R}^m,$$

where

$$x \mapsto \partial_{i_k} (\partial_{i_{k-1}} \cdots \partial_{i_1} f)(x).$$

The order of the indices records the order in which the derivatives are taken. Under suitable smoothness conditions, mixed partial derivatives commute.

0.2.2.11 Hessian

For a scalar-valued function $f: U \rightarrow \mathbb{R}$, the *Hessian* is

$$\nabla^2 f(x) := \begin{pmatrix} \partial_1 \partial_1 f(x) & \cdots & \partial_1 \partial_d f(x) \\ \vdots & \ddots & \vdots \\ \partial_d \partial_1 f(x) & \cdots & \partial_d \partial_d f(x) \end{pmatrix}.$$

The Hessian records the second-order curvature of a scalar-valued function.

Example 0.2.3 (Computing a Hessian). Let

$$f(x^{(1)}, x^{(2)}) = (x^{(1)})^2 + 3x^{(1)}x^{(2)} + 2(x^{(2)})^2.$$

The first derivatives are

$$\partial_1 f = 2x^{(1)} + 3x^{(2)}, \quad \partial_2 f = 3x^{(1)} + 4x^{(2)}.$$

Therefore, the Hessian is

$$\nabla^2 f(x) = \begin{pmatrix} 2 & 3 \\ 3 & 4 \end{pmatrix}.$$

0.2.2.12 Laplacian

For a scalar-valued function f , the *Laplacian* is

$$\Delta f := \text{tr}(\nabla^2 f).$$

Equivalently,

$$\Delta f = \partial_1^2 f + \partial_2^2 f + \cdots + \partial_d^2 f,$$

where $\partial_j^2 f := \partial_j \partial_j f$.

The Laplacian appears in diffusion, score models, graph Laplacians, manifold learning, and regularization.

0.2.2.13 Taylor's Theorem**0.2.2.14 One-Dimensional Case**

Let $f: I \subset \mathbb{R} \rightarrow \mathbb{R}$ and suppose $x + h \in I$. The order- k Taylor expansion used in these notes is

$$f(x + h) = f(x) + \partial f(x)h + \frac{1}{2}\partial^2 f(x)h^2 + \cdots + \frac{1}{k!}\partial^k f(x)h^k + o(h^k),$$

where

$$\lim_{h \rightarrow 0} \frac{o(h^k)}{|h|^k} = 0.$$

Thus, $o(h^k)$ denotes an error term that is negligible compared with $|h|^k$ as $h \rightarrow 0$.

An alternative version gives an explicit remainder. There exists a point $\xi \in [x, x + h]$ such that

$$f(x + h) = f(x) + \partial f(x)h + \cdots + \frac{1}{(k-1)!} \partial^{k-1} f(x)h^{k-1} + \frac{1}{k!} \partial^k f(\xi)h^k.$$

The derivatives through order $k-1$ are evaluated at x , while the final remainder uses $\partial^k f(\xi)$ for some ξ between x and $x + h$.

0.2.2.15 Second-Order Approximation

Taking $k = 2$ in the one-dimensional remainder formula gives

$$f(x + h) = f(x) + \partial f(x)h + \frac{1}{2} \partial^2 f(\xi)h^2, \quad \xi \in [x, x + h].$$

For a high-dimensional loss, the analogous approximation is

$$L(w + h) \approx L(w) + dL(w)h + \frac{1}{2} h^T \nabla^2 L(w)h.$$

The first-order term suggests a descent direction, while the second-order term controls the error and step size.

0.2.2.16 Gradient Descent from Taylor's Theorem

Take $h = -\eta \nabla L(w)$. Substitution into the second-order approximation gives

$$\begin{aligned} L(w + h) &\approx L(w) + dL(w)h + \frac{1}{2} h^T \nabla^2 L(w)h \\ &= L(w) - \eta \|\nabla L(w)\|_2^2 + \frac{\eta^2}{2} \nabla L(w)^T \nabla^2 L(w) \nabla L(w). \end{aligned}$$

The term $-\eta \|\nabla L(w)\|_2^2$ decreases the loss, while the order- η^2 term is the curvature penalty. Choosing η too large can therefore destroy descent.

0.2.3 Probability

0.2.3.1 Sample Spaces and Events

The *sample space* Ω is the set of all possible outcomes, and an *event* is a subset $A \subseteq \Omega$. A probability is a function assigning values to events,

$$\mathbb{P}: \mathcal{P}(\Omega) \rightarrow [0, 1],$$

such that

1. $\mathbb{P}(\Omega) = 1$;
2. $0 \leq \mathbb{P}(A) \leq 1$ for every $A \subseteq \Omega$;

3. if $A_1, A_2, \dots \subseteq \Omega$ are pairwise disjoint, then

$$\mathbb{P}\left(\bigcup_i A_i\right) = \sum_i \mathbb{P}(A_i).$$

The pair $(\Omega, \mathbb{P})$ is a *probability space*. Intuitively, the sample space lists all possible outcomes, while an event collects some of those outcomes.

Example 0.2.4 (One die roll). For one roll of a fair die,

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$

The event that the outcome is even is

$$A = \{2, 4, 6\}.$$

Since three of the six outcomes are even,

$$\mathbb{P}(A) = \frac{3}{6} = \frac{1}{2}.$$

0.2.3.2 Random Variables and Realizations

Random variables are functions defined on a probability space. We use capital letters $X, Y, \dots$ for random variables and lowercase letters $x, y, \dots$ for their realizations or observed values. The notation

$$[X = x]$$

denotes the event

$$\{\omega \in \Omega : X(\omega) = x\}.$$

In supervised learning, we often regard (X, Y) as a random input–output pair. A dataset

$$(x_1, y_1), \dots, (x_n, y_n)$$

consists of observed realizations of independent copies of (X, Y) . Thus, capital X, Y denote random variables before the data are observed, whereas lowercase x_i, y_i denote observed data points.